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Differential brainstem circuits mediating conditioned pain modulation induced analgesia and hyperalgesia: A cross-sectional, ultra-high field functional magnetic resonance imaging study

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ABSTRACT

Whilst preclinical studies have identified multiple brainstem sites that can modulate noxious information, the precise location and function of such sites in humans remains unresolved. One method to explore pain modulatory circuits in humans is through the use of the conditioned pain modulation paradigm. Conditioned pain modulation occurs when the intensity of one noxious stimulus is modulated by the application of a second noxious stimulus. Whilst conditioned pain modulation has been primarily used to explore pain inhibition, it can also evoke pain facilitation. Whether the same circuits are involved in both inhibition and facilitation remains unknown. We used ultra-high field functional magnetic resonance imaging to determine brainstem networks responsible for inhibitory and facilitatory conditioned pain modulation responses in 44 pain-free individuals. In 17 individuals pain decreased (inhibitors), in 11 individuals pain increased (facilitators), and 16 individuals displayed no pain intensity change. Analysis of brainstem signal changes revealed that whilst pain decreases were associated with signal intensity decreases in the subnucleus reticularis dorsalis, A5 cell group and locus coeruleus, pain increases were associated with no A5 signal change, but signal decreases in subnucleus reticularis dorsalis and locus coeruleus. Meanwhile, linear correlation analysis revealed midbrain periaqueductal gray matter signal changes that were negatively correlated and rostral ventromedial medulla signal changes that were positively correlated with pain intensity changes with both hypoalgesia and hyperalgesia. Our results suggest the existence of two brainstem networks whereby some regions respond in an on-off manner whereas others are linearly correlated with changes in pain intensity.

Perspective: This article presents the brainstem circuitry underpinning pain decreases and increases evoked by conditioned pain modulation paradigm. Since brainstem analgesic circuitry is thought to be dysfunctional in individuals with chronic pain, our data provides a platform for exploring such dysfunction.

Introduction

The descending pain modulatory system comprises multiple unique circuits that inhibit or facilitate dorsal horn or spinal trigeminal nucleus (SpV) neuronal responses eliciting analgesia and hyperalgesia.^{1–4} One validated method for evoking activity in this system and producing controlled experimental modulation is via concurrent application of two

distally applied noxious stimuli, i.e. a conditioned pain modulation (CPM) paradigm. This paradigm was typically used to explore whether an individual experiences an analgesic response when an initially painful ‘test’ stimulus is applied concurrent to a secondary noxious ‘conditioning’ stimulus, thus eliciting the so-called ‘pain inhibits pain’ phenomenon.⁵ However, it is now appreciated that in many individuals this experimental paradigm can instead evoke hyperalgesic responses.^{6,7}

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This is clinically relevant as individuals who exhibit facilitatory or non-responses, rather than inhibitory responses to the application of endogenous pain modulatory assessments such as the conditioned pain modulatory paradigm, are more likely to develop chronic pain post-injury.^{6,8}

CPM is believed to be underpinned by multiple mechanisms including diffuse noxious inhibitory control (DNIC). DNIC, quantified in anaesthetised animals as a significant reduction in stimulus-evoked dorsal horn neuronal activity upon application of a distally applied noxious conditioning stimulus,^{9,10} is mediated by supraspinal circuits. While lesions of well-described pain modulatory brainstem regions including the midbrain periaqueductal gray (PAG), rostral ventromedial medulla (RVM), locus coeruleus (LC) and parabrachial nucleus (PB) had little to no effect on the DNIC response,^{11–13} lesions of the medullary subnucleus reticularis dorsalis (SRD) produced a 50% reduction in DNIC responses.¹⁴ More recently, Bannister and colleagues have demonstrated that the A5 cell group originates the final descending control required for mediating the DNIC response,¹⁵ where reciprocity between the A5 and the LC is believed to underlie hyperalgesia.¹⁶

These findings appear consistent with a human functional magnetic resonance imaging (fMRI) study which found CPM-related signal changes in areas encompassing the SRD and dorsal pons with no observed changes in the PAG and RVM.¹⁷ While lesions of the PAG and RVM revealed no effect on DNIC responses, it was proposed that these regions may still play a regulatory role, rather than being critical for driving CPM or DNIC responses in humans and animals respectively.^{18,19} Moreover, the role of the A5 has not been explicitly explored in CPM in humans. Defining the precise brainstem regions responsible for pain modulation upon application of a CPM paradigm is essential for understanding their potential roles in the development and maintenance of chronic pain.

The aim of this investigation was to use ultra-high field (7 Tesla) fMRI to determine the brainstem network responsible for pain modulatory (inhibitory or facilitatory) responses in pain-free individuals during a CPM paradigm. We hypothesised that CPM-induced pain inhibition will be associated with increases in signal intensity of brainstem nuclei including the SRD, dorsal pons, A5 and the midbrain PAG whilst that of the LC and RVM will be associated with CPM-induced pain facilitation.

Methods

Participants

Forty-four pain-free individuals (23 males, 21 females – sex assigned at birth, mean±SEM age: 30.90±1.50) were recruited from the general population. Participants were eligible for the study if they were not suffering from any pain condition or any neurological condition, and fulfilled standard MRI inclusion criteria (not pregnant, claustrophobic, metal implants etc). Informed written consent was obtained from all participants and the investigation was conducted under the local Institutional Human Research Ethics Committees and is consistent with the Declaration of Helsinki.

All data acquisition was acquired at the Melbourne Brain Imaging Unit, Melbourne Australia between April 2019 and August 2023. Written informed consent was obtained from participants at the commencement of the study. Participants were provided with an emergency buzzer while inside the scanner so that they could stop the experiment at any time. In order to evaluate the necessary number of participants required for this study, an *a priori* power analysis was performed using G*Power 3.1.9.7 and results from a previous imaging study investigating brainstem changes during CPM analgesia.^{17,20} This revealed a total sample size of 40 would be necessary to detect similar effect sizes with 95% power ($d = 1.071$, $\alpha = 0.05$, power = 0.95).

Experimental design

Test stimulus

There is currently a lack of consistency in the methodology of experimentally inducing CPM, as well as a consensus on the location of the test and conditioning stimuli with the only specification that these be heterotopic.^{21,22} Currently, there is a shortage of investigation into the role of the trigeminal/orofacial nociceptive system in CPM and this holds clinical significance as individuals with chronic orofacial pain conditions such as migraine, headache and temporomandibular joint disorder have exhibited aberrant responses to CPM paradigms.^{23–25} The CPM protocol used in this investigation targets the orofacial nociceptive pathway via the application of a 3 cm x 3 cm MRI-compatible thermode (TSA-II, Medoc) delivering a pre-determined moderately painful temperature (test stimulus) to the right side of the lip of each participant 8 times.

Conditioning stimulus

Simultaneously, 0.5 mL of 5% hypertonic saline (conditioning stimulus) is injected into the tibialis anterior muscle of the right leg to induce sustained muscle pain. Intra-muscular hypertonic saline predominantly activates group III and IV afferent fibres which are primary nociceptor afferents.²⁶ It provides a standardized and reproducible pain response, with minimal activation of non-nociceptive fibres.

CPM protocol

Prior to entering the MRI scanner, the temperature that evoked a moderate level of pain, i.e. 4–5/10 on an 11-point Visual Analogue Scale was determined for each individual (VAS; '0' represents 'no pain' and '10' indicates 'worst pain imaginable'). Participants had a 3 cm x 3 cm MRI-compatible thermode (TSA-II, Medoc) placed on their left forearm delivering eight randomised temperatures ranging from 44°C to 48°C. The temperature that elicited a 5/10 pain intensity rating was used for subsequent MRI collection.

Each participant was then positioned supine in a 7 T MRI scanner (MAGNETOM, Siemen's Healthcare, Erlangen, Germany) with their head immobilised in a 32-channel receive head coil (Nova Medical, Wilmington MA, USA) with additional padding to prevent head movement. A three-dimensional T1-weighted anatomical image set of the whole brain was acquired (134 axial slices, repetition time: 5000 ms, echo time: 3.06 ms, raw voxel size: 0.73x0.73x0.73mm). Following this, the thermode was placed on the right side of the mouth covering the upper and lower lips of each participant and secured in place using a Velcro strap. A small needle attached to a 1 mL syringe filled with 5% saline was also inserted into the tibialis anterior muscle of the right leg. Two sets of fMRI images were collected covering the entire brain caudally to the level of the spinal cord. Each fMRI series consisted of 134 gradient-echo echo-planar fMRI volumes using blood oxygen level-dependent (BOLD) contrast collected over a period of 335 s (124 axial slices, repetition time 2500 ms; echo time 26 ms; voxel size: 1.0×1.0×1.2mm thick). Images were acquired in an interleaved collection pattern with a multi-band factor of four and an acceleration factor of three.

During the first (control) scan, a series of thermal stimuli were delivered via the thermode using the temperature pre-determined as eliciting a moderate pain intensity (test stimulus). The series of thermal stimuli commenced with a non-noxious baseline temperature of 32°C, followed by the delivery of the moderate heat over 15 s (2.5 s increase to the temperature, 10 s of maintenance, 2.5 s reduction to baseline). A 10 s inter-stimulus interval separated each stimulus (Fig. 1A). This was repeated for a total delivery of 8 noxious thermal stimulation periods. All participants rated their pain in real time using an MR-compatible button box to indicate on an online horizontal 10 cm visual analogue scale (VAS) their pain from each stimulus, ranging between 0 and 100, with 0 representing "no pain" and 100 "the worst pain imaginable." The

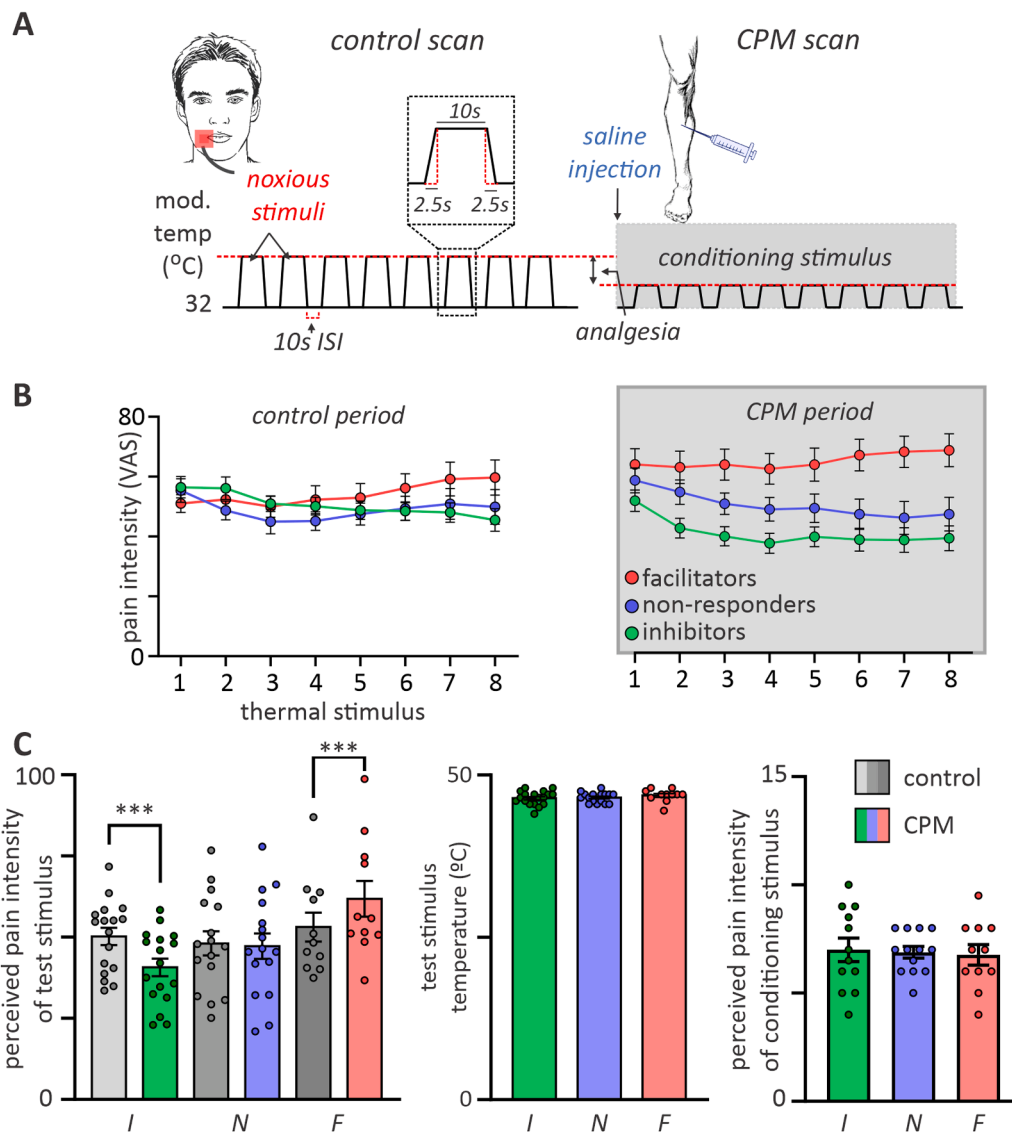


Fig. 1. Methods and psychophysics. A: The control scan involved the delivery of 8 noxious thermal stimuli to the right side of the lip (test stimulus: red). Each stimulus delivery comprised a 2.5 s increase from 32°C baseline to a pre-determined temperature that elicited a moderate pain intensity (44–48.5°C), 10 s of maintaining a moderately painful temperature, and a 2.5 s return to baseline (dashed pop-out box). Each stimulus period was separated by a 10 s interstimulus interval (ISI). The CPM scan involved the repetition of the test stimuli delivery as in the control scan, with the addition of a 0.5 mL bolus injection of 5% hypertonic saline into the tibialis anterior muscle of the right leg immediately prior to scan onset (saline injection point) to induce sustained muscle pain (conditioning stimulus: grey). B: Plots of mean±SEM perceived pain intensity during 8 thermal noxious stimuli during the control (left plot) and CPM scan (right plot) in inhibitors (green), non-responders (purple) and facilitators (red). Inhibitory responders displayed significantly lower pain ratings during the CPM period compared with non-responders and facilitators. Facilitators showed significantly higher pain ratings during the CPM period compared with non-responders and responders. C: Left plot shows mean±SEM pain intensity ratings of the test stimulus during the control period (grey) and CPM period for inhibitors (I: green), non-responders (N: purple), facilitators (F: red). Mean pain ratings for each participant in each group are indicated by corresponding coloured circles overlaid onto bar plot (***) $p < 0.001$. Middle plot shows mean±SEM temperatures eliciting moderate pain and used for the noxious thermal test stimuli. Right plot shows perceived pain intensity of the conditioning stimuli in I, N and F groups.

VAS scale was shown on a digital screen inside the scanner, with participants moving the slider left (towards 0) and right (towards 100). Participants were instructed to keep their eyes open, their minds relatively calm and continuously rate the pain intensity induced by the noxious thermal stimulation to the lip during this fMRI scan.

Subsequent to this first control series, a second fMRI (CPM) scan was acquired. Immediately prior to this fMRI scan, 0.5 mL of 5% hypertonic saline was injected into the right tibialis anterior muscle to induce sustained muscle pain (conditioning stimulus). The fMRI scan was then started, and a series of 8 noxious thermal stimulations were delivered identical to those delivered in the control scan (Fig. 1A). Like the preceding control scan, participants continuously rated their perceived pain

intensity of the lip only. At the conclusion of the fMRI scan, participants were also asked to provide an average rating of the intensity of the conditioning stimulus ('0' represents 'no pain' and '10' indicates 'worst pain imaginable').

Analysis

Psychophysics

Changes in pain intensity between control and CPM scans were determined by calculating the average pain intensity rating during all 8 noxious thermal stimuli during the control scan compared with the average of the 8 thermal stimuli during the CPM scan. Delineation of

those individuals that displayed a significant pain inhibitory response (*CPM inhibitors*), significant pain facilitatory response (*CPM facilitators*), and no change in response (*CPM non-responders*) was determined through permutation testing. As such, participant pain intensity ratings were permuted 10,000 times, sampled with replacement and mean differences were calculated from each permutation. An incidence curve was simulated and a significance threshold of $p < 0.05$ was set. That is, if a participant's mean difference in pain intensity ratings occurs with a probability of less than 5%, that individual has experienced a significant inhibitory or facilitatory effect upon application of the CPM paradigm. We have previously used this approach to designate participants into placebo inhibitors, non-responders and facilitators.²⁷ This resulted in 17 (39%) participants categorized as CPM inhibitors, 11 (25%) as CPM facilitators and 16 (36%) as CPM non-responders. To further verify these groupings, we also performed a two-way ANOVA to explore differences in pain intensities of the test stimulus between groups, within groups across the two scans (time variable), as well as an interaction effect ($p < 0.05$). To determine the influence of individual characteristics on CPM response status, significant differences between CPM inhibitor, facilitator and non-responder groups were determined for pain intensity rating variability (two-way ANOVA and Tukey's multiple comparisons test, $p < 0.05$), sex (chi-squared test, $p < 0.05$), as well as age, temperature of the test stimulus, and intensity of the conditioning stimulus (two-sample t-test, $p < 0.05$).

fMRI preprocessing

All images were processed using Statistical Parametric Mapping 12²⁸ and custom software. All fMRI image sets were slice timing and motion corrected, ensuring head movements did not exceed 1 mm in the X, Y or Z planes, or 0.5 radians in the roll, yaw or tilt rotations. Linear modelling of parameters (LMRP) detrending²⁹ was used to remove any effects of movement, and a Dynamic Retrospective Filtering (DRIFTER) toolbox corrected for physiological cardiac (frequency band of 60–120 beats per minute ± 1 harmonic) and respiratory (frequency band of 8–25 breaths per minute ± 1 harmonic) noise from fMRI signal.³⁰ fMRI image sets were then linearly detrended to remove any global signal drifts. The Synthetic BOLD Distortion Correction (SynBOLD DisCO) toolbox was used to correct for spatial distortion resulting from susceptibility to inhomogeneities of the magnetic field that can occur during echo-planar imaging at ultra-high field resolutions, as well as artifacts arising from facial stimuli.³¹ All functional images were then co-registered to their respective T1-weighted anatomical images and normalised to Montreal Neurological Institute (MNI) template space. With focus on the brainstem circuitry involved in CPM, the Spatially Unbiased Infratentorial Template (SUIT) toolbox³² was used on fMRI and T1 image sets to isolate the brainstem and cerebellum. This software is specifically designed to provide more accurate spatial normalisation of the brainstem and cerebellum and we have previously used this toolbox to accurately map signal changes during noxious stimuli in discrete columns of the midbrain periaqueductal gray matter and specific hypothalamic nuclei.^{33,34} These “brainstem-only” images were then spatially normalised to the brainstem and cerebellum template in MNI space, resliced into 1 mm isotropic voxels and spatially smoothed using a 2 mm full-width-at-half maximum (FWHM) Gaussian filter. This accounted for the inter-individual variability in brainstem anatomy and permitted rigorous brainstem-specific analysis.

fMRI statistical analysis - signal intensity changes

Significant changes in signal intensity during thermal noxious stimuli were determined with the use of a repeating box-car model with the noxious stimulus periods assigned values of ‘1’, and the inter-stimulus-interval values of ‘0’. This was convolved with a canonical haemodynamic response function to account for the delay in blood oxygen level dependent (BOLD) response. For each participant, brain activation β maps were created for the control and CPM scans separately and then difference maps were constructed by the subtraction of the control β

map from the CPM β map. The resultant images represent the change in signal intensity between the two functional scans in each individual.

For the following analyses, two samples were defined to explicitly explore the brainstem signal intensity changes associated with CPM-induced pain inhibition and facilitation. The first comprised inhibitors and non-responders, and the second, facilitators and non-responders. In both samples, non-responders acted as baselines, expressing no distinct qualities of facilitation or inhibition. Meanwhile, the inhibitory and facilitatory responders exhibited the characteristic hypo- and hyperalgesic responses that distinguish the two samples.

Four analyses were conducted using the constructed difference maps to determine brainstem-specific signal intensity changes during CPM-induced inhibitory and facilitatory responses. The first two analyses consisted of two-sample voxel-by-voxel, random effects analyses to determine signal intensity change differences between CPM inhibitors and non-responders, and CPM facilitators and non-responders ($p < 0.005$ uncorrected).

Additionally, exploratory conjunction analyses were conducted to identify potential overlapping brainstem regions exhibiting significant signal intensity changes across the two-sample analyses (inhibitors versus non-responders and facilitators versus non-responders). To examine group differences in brainstem signal intensity changes during CPM, we performed a one-way ANOVA on extracted β values for each identified brainstem cluster, followed by Tukey's post-hoc multiple comparisons. Significant pairwise differences were observed between inhibitors and non-responders, facilitators and non-responders, and inhibitors and facilitators ($p < 0.05$, FWE-corrected).

The following two analyses consisted of two one-sample linear regression analyses. The first comprising inhibitors and non-responders and the second comprising facilitators and non-responders to determine significant relationships between brainstem signal intensity changes and CPM-evoked pain inhibition and pain facilitation, respectively ($p < 0.005$, uncorrected). The more ‘positive’ the CPM response, the greater the hypoalgesia and the more ‘negative’, the greater the hyperalgesia. For all two-sample and linear regression analyses, a cluster extent threshold of 4 contiguous voxels was used to minimise the possibility for Type 1 errors. To investigate the exclusivity of the observed correlated signal intensity changes to CPM hypoalgesia (inhibitory and non-responder) and CPM hyperalgesia (facilitatory and non-responder), a comparison of correlation coefficients was conducted. In RStudio, a multiple linear regression model was constructed with the independent variables X_1 being CPM ability, and X_2 group (*inhibitory*: inhibitors and non-responders, *facilitatory*: facilitators and non-responders). The interaction term X_1X_2 was defined as the multiplication of the two independent variables to investigate the effect of group on the relationship between the observed signal intensity changes in identified brainstem clusters and CPM ability.³⁵ The mean differences in slopes of the regression were calculated and p-values determined, corrected with Tukey's multiple comparisons.

Since we hypothesised that regions of the PAG, RVM, LC, A5, PB, SRD modulate incoming noxious information directly or indirectly at the level of the SpV, we restricted the analyses to these nuclei, which were identified using a brainstem atlas (Supplementary Figure 1).³⁶ Small volume corrections (SVC) were applied in all analyses using spheres of 2 mm radii centred at a priori coordinates for each nucleus derived from previous relevant neuroimaging publications^{17,34,37,38} and brainstem atlases.^{36,39} A threshold $p < 0.05$ family-wise error (FWE) corrected was set after small volume correction. Nuclei that exhibited signal intensity changes within these spheres and survived SVC were deemed valid and therefore reported. Signal intensity changes (β values) for each significant cluster in all analyses were extracted from the difference β maps of all three response groups to explore the signal changes in more detail.

PAG-specific analyses

Significant negative relationships were observed between CPM inhibitory ability and the PAG at rostrocaudal level –5, as well as CPM

hyperalgesic ability and the PAG at z-level -7. As it is known that the PAG contains four functionally discrete longitudinal columns,⁴⁰ we finally ran a PAG-specific analysis to determine the precise PAG columns in which the signal intensity changes occurred. Using MRICron, masks encompassing the dorsomedial (dmPAG), dorsolateral (dlPAG), lateral (PAG) and ventrolateral PAG (vlPAG) columns on the left and right were manually drawn at 1 mm intervals throughout its rostro-caudal extent using an average anatomical image as a template (z co-ordinate (MNI space) = -3 to -11). Columns were derived from a brainstem atlas,³⁶ diffusion based segmentation⁴¹ and preclinical studies of PAG columnar organisation⁴⁰ and are consistent with the columns used previously to explore regional PAG signal changes during placebo analgesia and nocebo hyperalgesia.^{34,37} The β values from signal intensity change difference β maps were extracted from CPM inhibitors, facilitators and non-responders for each mask at the rostrocaudal level -5 and -7 and linear regression analyses of CPM inhibitors and non-responders, and facilitators and non-responders were performed to determine the exact columns that significantly correlated with CPM hypo- and hyperalgesia, respectively.

To extend the PAG-specific analysis, two analyses were conducted to compare the significance of correlations between signal intensity change and CPM abilities of different PAG columns. A within-group analysis explored whether the signal intensity changes of particular PAG columns were more correlated with CPM ability than others. A between-group analysis explored differences in the correlations of PAG columnar signal intensity changes with CPM ability between the defined inhibitory and non-responder and facilitatory and non-responder groups. For both analyses, the difference between the two correlation coefficients were computed via RStudio, fitting an interaction term (X_1X_2) to a multiple linear regression model where X_1 represents the independent variable CPM ability, and X_2 PAG column. The correlations between the signal intensity changes of each PAG column with CPM ability were compared between and within-groups at rostrocaudal levels -5 and -7. Mean differences between correlations were calculated and p-values determined, corrected with Tukey's multiple comparisons ($p < 0.05$ FWE).

Results

Psychophysics

As mentioned above, permutation testing resulted in 17 (39%) participants categorized as CPM inhibitors, 11 (25%) as CPM facilitators and 16 (36%) as CPM non-responders. We verified these groupings, finding a significant group ($F(2,42)=4.750$, $p=0.0138$) and a group x scan interaction effect ($F(2,42)=42.67$, $p<0.0001$). That is, the difference in pain intensities between scans is dependent on the group (inhibitors, non-responders or facilitators). During the CPM scan, facilitators exhibited significantly higher pain ratings than inhibitors and non-responders (mean±SEM VAS: inhibitors 38.52±3.36; non-responders 46.49±3.99; facilitators 64.80±4.90, $p<0.005$) (Figs. 1B

Table 1
Mean±SEM age (years) of inhibitors, non-responders and facilitators and p-value extracted from one-way ANOVA. Ratios and proportions (%) of males (M) and females (F) are displayed for inhibitors, non-responders and facilitators, corresponding p-values were extracted from chi-squared tests conducted on raw numbers of males and females, and percentages.

	inhibitors (n=17)	non-responders (n=16)	facilitators (n=11)	p-value
age (mean ±SEM years)	30.37±2.20	32.58±3.00	29.34±2.00	0.69
sex, M/F	9/8	10/6	4/7	0.41
males (%)	53	63	36	
females (%)	47	37	64	> 0.05

and 1C, Table 1). Assessing within-group differences in pain intensities, inhibitors had significantly lower ratings during the CPM scan compared to the control (hypoalgesia) (mean±SEM VAS pain intensity (out of 100): control 49.72±2.57; CPM 38.52±3.36, $p<0.0001$), facilitators had significantly increased ratings (hyperalgesia) (control 55.19±4.23; CPM 64.80±4.90, $p<0.0001$), while non-responders exhibited no change (control 46.63±3.56; CPM 46.49±3.99, $p=0.92$).

Pain intensity ratings of each of the 8 noxious thermal stimuli during the control and CPM scans for each participant are plotted in Supplementary Figure 1. It can be seen that inhibitory participants exhibited decreased pain intensities during almost all thermal noxious stimuli during the CPM compared with the control scan (Supplementary Figure 2A). Whilst non-responders displayed no obvious response patterns (Supplementary Figure 2B), facilitators demonstrated hyperalgesic responses over almost all 8 thermal stimuli during the CPM compared with control scan (Supplementary Figure 2C). With respect to pain intensity rating variabilities, we found no significant main effect of scan ($F(1,84)=2.68$, $p=0.11$), of group ($F(2,84)=2.37$, $p=0.10$) or an interaction effect ($F(2,84)=0.10$, $p=0.90$). Additional Tukey's multiple comparisons test confirmed no significant differences between the three groups in both the control (mean SD: inhibitors 7.65; non-responders 9.17; facilitators 7.11, $p>0.05$) and CPM scan (mean SD: inhibitors 6.35; non-responders 8.002; facilitators 4.88, $p>0.05$). No significant difference was observed between the two scans in the three groups also ($p>0.05$) (see Supplementary Figure 3, Supplementary Table 1).

Despite significant pain intensity responses between groups, a one-way ANOVA with Tukey's multiple comparisons revealed no significant difference in the temperatures perceived as moderately painful between the three groups (mean±SEM temperature (°C): inhibitors 46.40±0.25; non-responders 46.40±0.23; facilitators 47.10±0.18, $F(2,42)=2.29$, $p=0.11$) (Fig. 1C). Similarly, no significant difference in the perceived intensity of the conditioning stimulus was observed between the three groups (mean±SEM VAS score (out of 100): inhibitors 64.60±6.012; non-responders 69.60±2.42; facilitators 66.80±4.37, $F(2,34)=0.23$, $p=0.80$). In addition, age was not significantly different between the three groups (mean±SEM years: inhibitors 30.37±2.20; non-responders 32.58±3.00; facilitators 31.17±2.49, $F(2,42)=0.37$, $p=0.69$) and there was no significant relationship between age and CPM ability (%change in VAS score) ($r=-0.05$, $p=0.74$). There were also no significant sex differences in the three groups (inhibitors 50% males, 50% females; non-responders: 36% males, 64% females; facilitators 62% males, 36% females, $\chi^2(1,45) = \chi^2(1,45)=1.80$, $p=0.41$), or sex differences in CPM ability (mean±SEM % change in VAS: males -6.04±3.42; females -5.17±5.53, $p=0.90$).

Furthermore, assessment of pain intensity ratings of the test stimuli during the initial control fMRI scan revealed no significant difference between all three groups (mean±SEM VAS pain intensity (out of 100) during control scan: inhibitors 49.72±2.67; non-responders 46.63±3.56; facilitators 55.19±4.23, $F(2,42)=1.393$, $p=0.26$). One-way ANOVA revealed a significant group effect during the CPM scan with facilitators expectedly showing significantly greater pain intensity of the thermal stimulus compared to inhibitors and non-responders, however no significant difference between inhibitors and non-responders (mean±SEM VAS pain intensity (out of 100) during CPM scan: inhibitors 38.52±3.36; non-responders 46.49±3.99; facilitators 64.80±4.90, $F(2,42)=10.07$, $p=0.0003$). Despite this, inhibitors and facilitators demonstrated significant hypoalgesia and hyperalgesia during the CPM scan while non-responders exhibited no significant change in pain intensity between the two scans (mean±SEM change in VAS pain intensity: inhibitors -11.20±1.97; non-responders -0.15±0.86; facilitators 9.61±1.14, $F(2,42)=42.67$, $p<0.0001$) (Figs. 1B and 1C).

Signal intensity changes

CPM inhibitors versus non-responders

Investigations of signal intensity change differences (CPM-control)

between inhibitory and non-responder groups revealed greater signal intensity change differences in inhibitors in the contralateral substantia nigra (SN), A5 cell group, rostral ventromedial medulla (RVM) and ipsilateral spinal trigeminal nucleus (SpV). That is, these nuclei exhibited increased signal intensity during CPM compared to control scans. In

contrast, signal intensity was significantly lower in inhibitors in the contralateral locus coeruleus (LC) and ipsilateral SRD (Fig. 2A, Table 2). Relative to non-responders, inhibitors exhibited reduced signal intensity in these nuclei during CPM compared to the control scan.

Post-hoc ANOVA with Tukey's multiple comparisons emphasised the

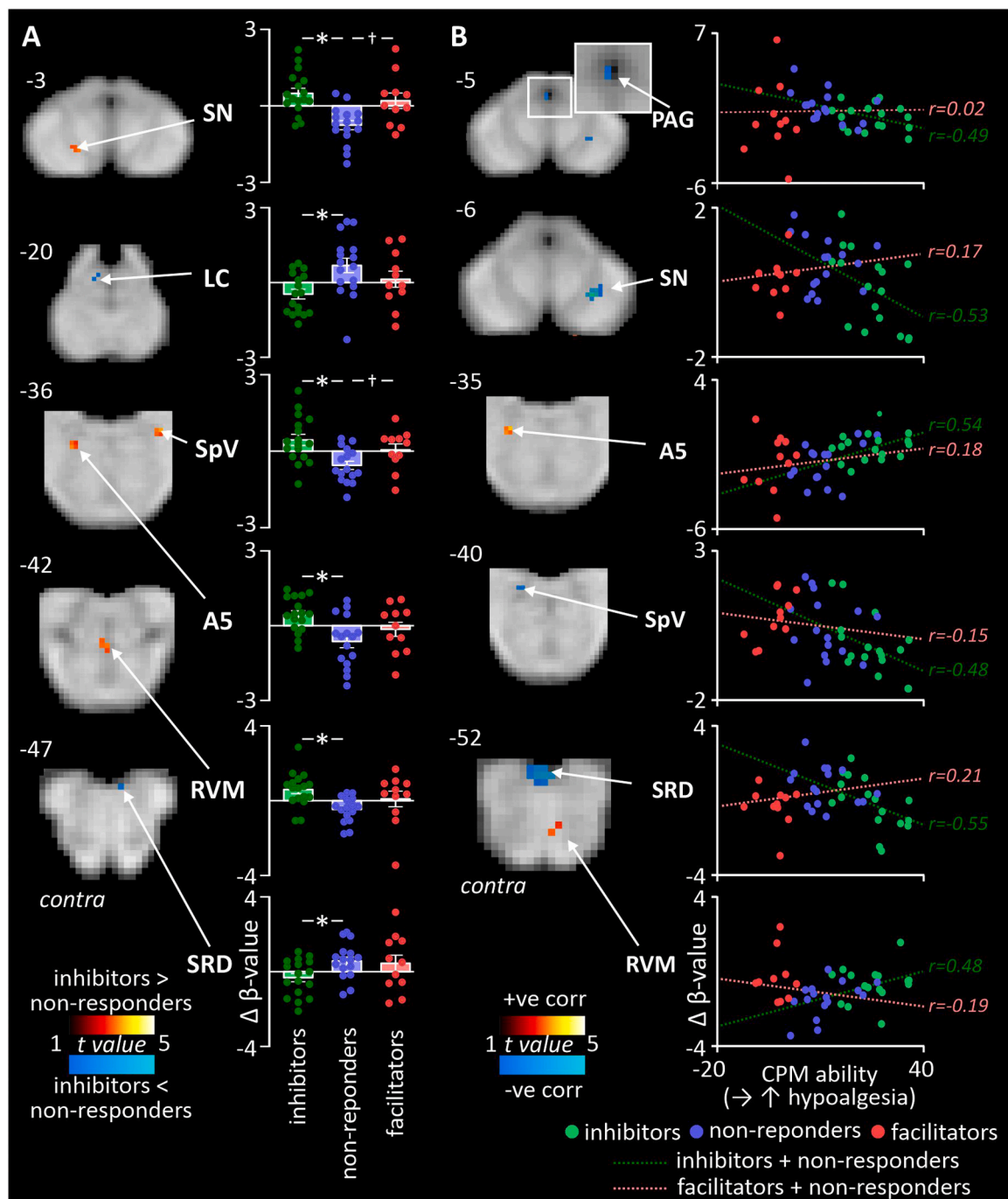


Fig. 2. Signal intensity changes associated with CPM hypoalgesia. A: Clusters with significant signal change differences between control and conditioned pain modulation (CPM) scans in inhibitors and non-responders overlaid onto axial slices of a mean brainstem T1-weighted anatomical image. Signal intensity changes greater in inhibitory responders than non-responders are indicated by the hot colour scale and changes lower in inhibitors than non-responders by the cool colour scale. Plots of mean±SEM signal intensity change differences (CPM-control) of each cluster extracted from inhibitors, non-responders and facilitators are shown to the right. B: Clusters with significant signal change differences between control and CPM scans (CPM-control) in inhibitory and non-responder individuals correlated with CPM ability (% pain intensity change) overlaid onto axial slices of a mean brainstem T1-weighted anatomical image. The more positive the value the greater the decrease in pain intensity (hypoalgesia). Positive correlations are indicated by the hot colour scale and negative correlations by the cool colour scale. Plots of mean signal intensity change differences against pain intensity change are shown to the right for both the inhibitory + non-responder as well as for the facilitatory + non-responder groups. SN: substantia nigra; LC: locus coeruleus; SpV: spinal trigeminal nucleus; RVM: rostral ventromedial medulla; SRD: subnucleus reticularis dorsalis; PAG: midbrain periaqueductal gray; contra: contralateral. * $p < 0.05$ inhibitors versus non-responders † $p < 0.05$ facilitators versus non-responders, two-sample t-test.

Table 2

Location, cluster size and t values of regions which were significantly different between control and conditioned pain modulation (CPM) scans during inhibitory versus non-responders two-group and correlation analyses. Mean±SEM β values changes (CPM - control scans) are also shown extracted from all three groups: inhibitor, non-responder and facilitator groups. MNI: Montreal Neurological Institute space; SN: substantia nigra; SpV: spinal trigeminal nucleus; RVM: rostral ventromedial medulla; SRD: subnucleus reticularis dorsalis; PAG: midbrain periaqueductal gray matter; contra: contralateral; ipsi: ipsilateral. * $p < 0.05$ inhibitors versus non-responders and inhibitors + non-responders correlation voxel-by-voxel analyses. † $p < 0.05$ facilitators versus non-responders, two-sample t-test.

	MNI coordinates			cluster size	t value	mean±SEM β value changes(CPM -control scans)		
	X	Y	Z			inhibitors	non-responders	facilitators
<i>inhibitors > non-responders</i>								
contra SN	-7	-18	-6	32	3.29	0.49±0.19*	-0.75±0.17	0.20±0.29 [†]
contra A5	-8	-35	-36	12	3.27	0.44±0.16*	-0.63±0.24	-0.14±0.27
ipsi SpV	13	-39	-36	9	3.40	0.44±0.21*	-0.55±0.16	0.05±0.22 [†]
	0	-32	-43	10	3.49	0.60±0.21*	-0.47±0.16	0.09±0.40
RVM	0	-26	-41	4	3.05	0.47±0.13*	-0.51±0.28	-0.09±0.29
<i>inhibitors < non-responders</i>								
contra LC	-3	-34	-19	4	3.38	-0.46±0.19*	0.69±0.28	0.13±0.31
ipsi SRD	3	-43	-47	1	2.90	-0.32±0.21*	0.57±0.21	0.45±0.41
						inhibitors + non-responders	facilitors + non-responders	difference between r-values
<i>positive correlations with pain intensity change</i>								
contra A5	-8	-35	0.18	7	3.60	r=-0.54*	r=0.18, p=0.38	p=0.12
RVM	2	-36	0.09	3	3.03	r=0.48*	r=-0.19, p=0.36	p=0.009*
<i>negative correlations with pain intensity change</i>								
contra PAG	0	-29	-5	3	2.93	r=-0.49*	r=0.02, p=0.90	p=0.04*
contra SpV	-6	-40	-41	3	2.92	r=-0.48*	r=-0.15, p=0.47	p=0.17
ipsi SN	11	-18	-6	19	4.22	r=-0.53*	r=0.17, p=0.41	p=0.0002*
RVM	2	-43	-46	4	3.27	r=-0.48*	r=-0.19, p=0.36	p=0.004*
SRD	1	-44	-51	20	3.93	r=-0.55*	r=0.21, p=0.30	p=0.002*

association between these findings and inhibitory CPM profiles (Supplementary Table 2). That is for all clusters, signal intensity change differences were significantly different between the inhibitor and non-responder groups but not significantly different between the facilitator and non-responder group. In addition, signal intensity change differences were greater in facilitators than non-responders in the contralateral SN and ipsilateral SpV (Fig. 2A, Table 2). However, the mean difference between facilitators and non-responders did not survive statistical significance in the post-hoc ANOVA with Tukey's multiple comparisons for any cluster (Supplementary Table 2).

CPM inhibitors and non-responders correlated with pain intensity change

Signal intensity change differences were significantly correlated with pain intensity changes in multiple regions (Fig. 2B, Table 2). Significant positive relationships occurred in the contralateral A5 and RVM i.e. the greater the pain reduction the greater the signal intensity change increase. In contrast, significant negative correlations occurred in the contralateral PAG and ipsilateral SN, RVM and SRD. Extraction of β values from facilitators for each cluster revealed a significant relationship between the ipsilateral SN with CPM hyperalgesic ability (Table 2). Comparison of correlation coefficients revealed that the RVM displayed a significantly stronger correlation with CPM ability in the inhibitory and non-responder compared to the facilitatory and non-responder group where it exhibited a non-significant negative relationship. Meanwhile, SN, RVM and SRD exhibited a negative relationship with CPM inhibitory ability that was significantly different to and stronger than the positive relationship observed in the facilitatory and non-responder group (Table 2).

CPM facilitators versus non-responders

Investigations of signal intensity change differences (CPM-control) between facilitators and non-responder groups revealed greater signal intensity change differences in facilitators in the ipsilateral lateral parabrachial nucleus (IPB). In other words, relative to non-responders, facilitators exhibited greater signal intensity during CPM compared to control scan in this nucleus. Reduced signal intensity was observed in the contralateral LC, ipsilateral SRD and bilateral SN, that is, lower signal intensity in CPM compared to control relative to non-responders (Fig. 3A, Table 3). These differences were consistent with post-hoc ANOVA analysis with the exception of the contralateral LC that did

not survive correction with Tukey's multiple comparisons (Supplementary Table 3).

In addition, β values extracted from each cluster for the CPM inhibitory group revealed reduced signal intensity change differences in the IPB and contralateral LC and SN compared to the facilitators, as well as the SRD compared to non-responders. On the other hand, signal intensity in the ipsilateral pontine nuclei and SN was greater compared to facilitators (Fig. 3A, Table 3). Post-hoc ANOVA with Tukey's multiple comparisons confirmed these consistent differences between inhibitors and facilitators (Supplementary Table 3).

CPM facilitators and non-responders correlated with pain intensity change

Signal intensity change differences were significantly correlated with pain intensity changes in multiple regions (Fig. 3B, Table 3). Significant positive relationships occurred in the contralateral region encompassing the LC, ipsilateral SpV, bilateral SN and RVM. Negative relationships were observed with the contralateral SpV and ipsilateral IPB and PAG. Extraction of β values from inhibitors for each cluster revealed no significant relationships between signal intensity differences and CPM inhibitory ability in any of the clusters. Comparison of correlation coefficients revealed that the bilateral SN and RVM exhibited a significantly stronger positive correlation with CPM ability while the ipsilateral PAG and PB exhibited a significantly more negative relationship with CPM ability in the facilitatory and non-responder compared the inhibitory and non-responder group (Table 3).

Midbrain periaqueductal gray matter column changes

Since we observed that the correlations result in the PAG appeared to be in different locations for the inhibitory and facilitatory group analysis, we conducted a columnar analysis. At -5 and -7 , β values from each PAG column (dorsomedial, dorsolateral, lateral and ventrolateral) were extracted and correlations with pain intensity changes made. We found that at z-level -5 , signal intensity was correlated with pain intensity change only in the inhibitory group analysis and only within the contralateral lateral PAG column (Fig. 4, Table 4). In contrast, at z-level -7 , signal intensity was correlated with pain intensity change only in the facilitatory group analysis and bilaterally in the ventrolateral and lateral PAG columns.

Within-group analyses of the inhibitory group revealed no significant difference between any of the columns at z-levels -5 and -7 . The

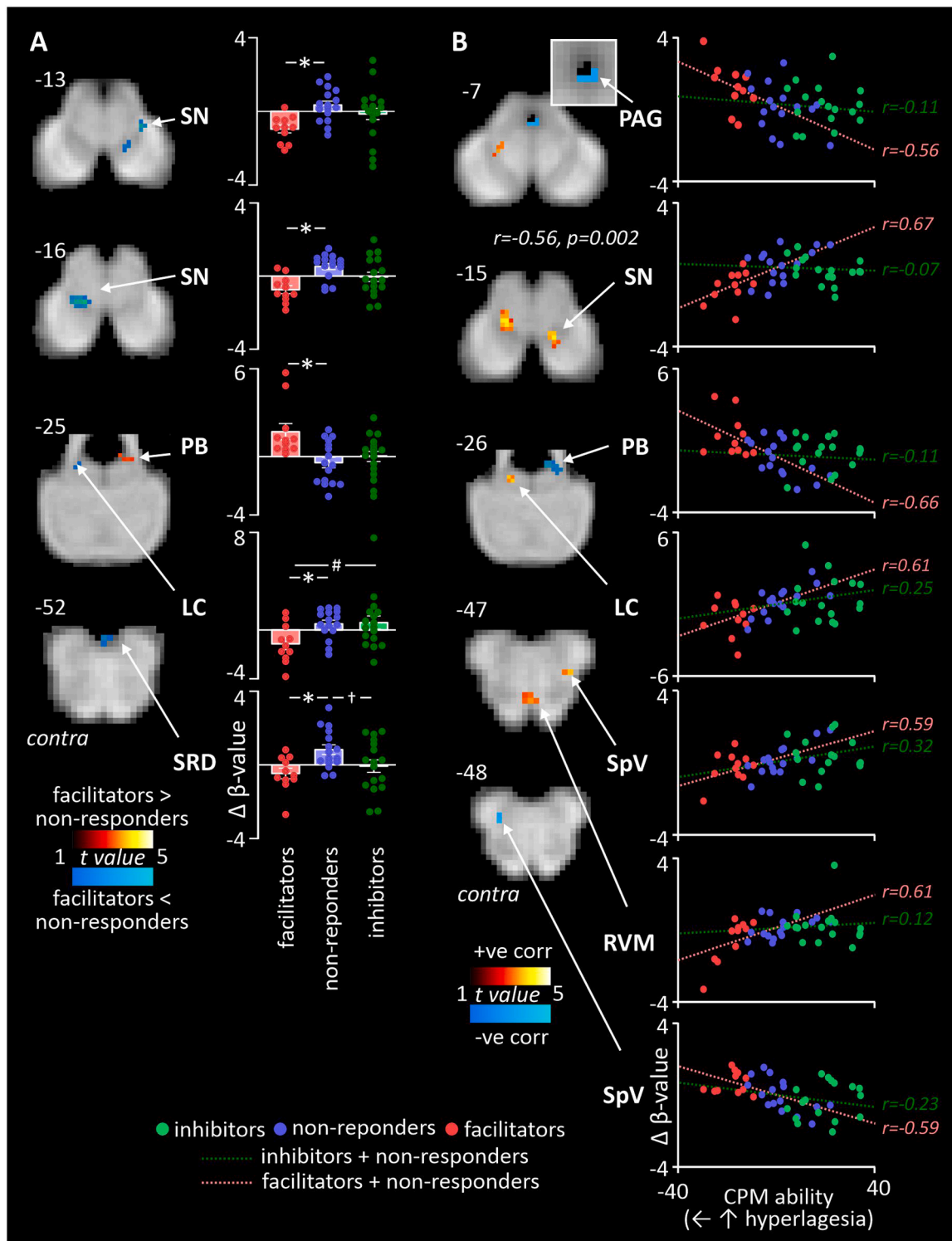


Fig. 3. Signal intensity changes associated with CPM hyperalgesia. **A:** Clusters with significant signal change differences between control and conditioned pain modulation (CPM) scans in facilitators and non-responders overlaid onto axial slices of a mean brainstem T1-weighted anatomical image. Signal intensity changes greater in facilitators compared to non-responders denoted by the hot colour scale and changes lower in facilitators than non-responders indicated by the cool colour scale. Plots of mean $\pm$ SEM signal intensity change differences (CPM-control) of each cluster extracted from facilitators, non-responders and inhibitors are shown on the right. **B:** Clusters with significant signal change differences between control and CPM scans (CPM-control) in facilitators and non-responder individuals correlated with change in CPM ability (% pain intensity change) overlaid onto axial slices of a mean brainstem T1-weighted anatomical image. The more negative the value, the greater the increase in pain intensity (hyperalgesia). Positive correlations are indicated by the hot colour scale and negative correlations by the cool colour scale. Plots of mean signal intensity change differences against pain intensity change are shown to the right for both the inhibitory + non-responder as well as for the facilitatory + non-responder groups. IPB: lateral parabrachial nucleus; SRD: subnucleus reticularis dorsalis; LC: locus coeruleus; SpV: spinal trigeminal nucleus; PAG: midbrain periaqueductal gray matter; RVM: rostral ventromedial medulla; contra: contralateral. * $p < 0.05$ facilitators versus non-responders and. † $p < 0.05$ inhibitors versus non-responders, two-sample t-test. # $p < 0.05$ facilitators versus inhibitors, two-sample t-test.

Table 3

Location, cluster size and t values of regions which were significantly different between control and conditioned pain modulation (CPM) scans during facilitatory versus non-responder two-group and correlation analyses. Mean±SEM β values changes (CPM -control scans) are also shown extracted from all three groups: facilitators, non-responder and inhibitor groups. MNI: Montreal Neurological Institute space; IPB: lateral parabrachial nucleus; SRD: subnucleus reticularis dorsalis; SN: substantia nigra; LC: locus coeruleus; SpV: spinal trigeminal nucleus; RVM: rostral ventromedial medulla; PAG: midbrain periaqueductal gray matter; contra: contralateral; ipsi: ipsilateral. * $p < 0.05$ facilitators versus non-responders and facilitators + non-responders correlation voxel-by-voxel analyses. † $p < 0.05$ inhibitors versus non-responders, two-sample t-test. # $p < 0.05$ facilitators versus inhibitors, two-sample t-test.

	MNI coordinates			cluster size	t value	mean±SEM β value changes(CPM -control scans)		
	X	Y	Z			facilitators	non-responders	inhibitors
<i>facilitators > non-responders</i>								
ipsi IPB	5	-40	-25	4	2.92	1.70±0.53*	-0.42±0.35	0.004±0.33 [#]
<i>facilitators < non-responders</i>								
contra SN	-4	-26	-16	25	4.33	-0.74±0.21*	0.53±0.17	-0.05±0.25
	-10	-24	-6	4	2.87	-1.08±0.36*	0.45±0.33	-0.34±0.25
contra LC	-5	-37	-25	3	2.97	-1.13±0.46*	0.51±0.29	0.61±0.53 [#]
ipsi SRD	1	-45	-52	4	3.27	-0.46±0.27*	0.83±0.26	-0.06±0.33 [†]
ipsi SN	10	-27	-13	9	3.98	-0.95±0.20*	0.35±0.20	-0.14±0.30
	6	-22	-13	7	3.11	-0.97±0.31*	0.50±0.28	0.12±0.25 [#]
						facilitators + non-responders	inhibitors + non-responders	difference between r-values
<i>positive correlations with pain intensity change</i>								
contra SN	-5	-27	-15	32	4.02	r=0.68*	r=-0.28, p=0.12	p=0.048*
	-9	-22	-6	69	3.80	r=0.82*	r=0.10, p=0.55	p=0.0001*
contra LC	-4	-36	-26	5	3.68	r=0.61*	r=0.25, p=0.16	p=0.10
ipsi SN	7	-21	-13	39	4.32	r=0.67*	r=-0.07, p=0.68	p=0.001*
ipsi SpV	8	-38	-47	6	3.63	r=0.59*	r=0.32, p=0.07	p=0.21
RVM	1	-33	-47	20	3.35	r=0.61*	r=0.12, p=0.51	p=0.03*
<i>negative correlations with pain intensity change</i>								
contra SpV	-5	-40	-49	4	3.24	r=-0.59*	r=-0.23, p=0.20	p=0.11
ipsi PAG	0	-30	-7	4	3.21	r=-0.56*	r=-0.11, p=0.53	p=0.06
ipsi IPB	5	-40	-25	36	4.25	r=-0.66*	r=-0.11, p=0.55	p=0.01*

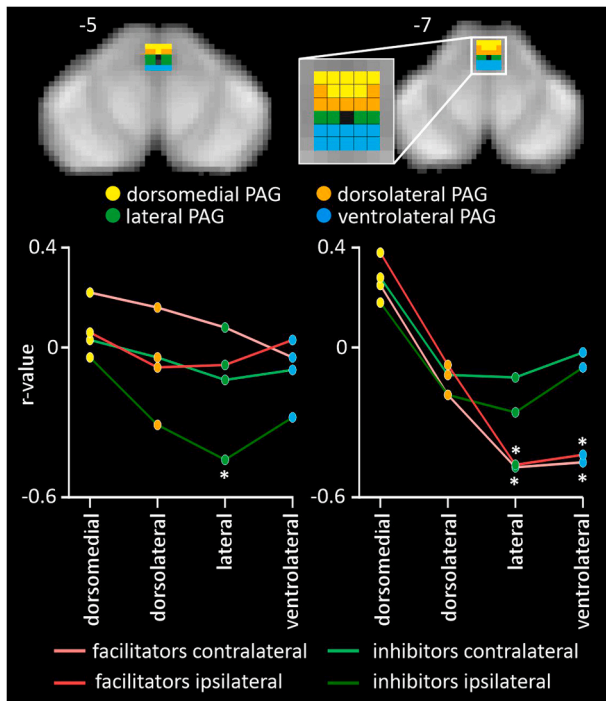


Fig. 4. Differential activation of the periaqueductal gray columns. Outlines of individual periaqueductal gray (PAG) columns masks overlaid onto axial slices of a mean T1-weighted anatomical image located at rostrocaudal level -5 and -7 in Montreal Neurological Institute (MNI) space (top). Plots of columns that are significantly correlated with conditioned pain modulation ability (bottom). * $p < 0.05$; contra: contralateral (bold lines); ipsi: ipsilateral (pastel lines).

strength of correlation of the PAG columns were also compared within the facilitatory group. Like the inhibitory group, no significant differences were observed between any of the PAG columns in terms of correlation with CPM facilitatory ability at z-level -5 and. A significant

Table 4

R-values and p-values of signal intensity change differences correlating with conditioned pain modulatory (CPM) ability in inhibitor, non-responder, and facilitator groups. for each midbrain periaqueductal gray matter (PAG) column at rostrocaudal levels -5 and -7 ipsi: ipsilateral; contra: contralateral. * $p < 0.05$.

	inhibitors + non-responders	facilitators + non-responders
rostrocaudal level -5		
ipsilateral dorsomedial PAG	$r = 0.03, p = 0.89$	$r = 0.22, p = 0.26$
contralateral dorsomedial PAG	$r = -0.04, p = 0.84$	$r = 0.06, p = 0.76$
ipsilateral dorsolateral PAG	$r = -0.04, p = 0.83$	$r = 0.16, p = 0.41$
contralateral dorsolateral PAG	$r = -0.31, p = 0.08$	$r = -0.08, p = 0.71$
ipsilateral lateral PAG	$r = -0.13, p = 0.46$	$r = 0.08, p = 0.68$
contralateral lateral PAG	$r = -0.45, p = 0.008^*$	$r = -0.07, p = 0.73$
ipsilateral ventrolateral PAG	$r = -0.09, p = 0.62$	$r = -0.04, p = 0.83$
contralateral ventrolateral PAG	$r = -0.28, p = 0.12$	$r = 0.03, p = 0.67$
rostrocaudal level -7		
ipsilateral dorsomedial PAG	$r = 0.28, p = 0.11$	$r = 0.25, p = 0.20$
contralateral dorsomedial PAG	$r = 0.18, p = 0.31$	$r = 0.38, p = 0.05$
ipsilateral dorsolateral PAG	$r = -0.11, p = 0.54$	$r = -0.19, p = 0.34$
contralateral dorsolateral PAG	$r = -0.19, p = 0.28$	$r = -0.07, p = 0.72$
ipsilateral lateral PAG	$r = -0.12, p = 0.51$	$r = -0.48, p = 0.01^*$
contralateral lateral PAG	$r = -0.26, p = 0.15$	$r = -0.47, p = 0.01^*$
ipsilateral ventrolateral PAG	$r = -0.02, p = 0.90$	$r = -0.46, p = 0.02^*$
contralateral ventrolateral PAG	$r = -0.08, p = 0.67$	$r = -0.43, p = 0.03^*$

difference was observed between the correlations of the ipsilateral dmPAG and bilateral lPAG and vPAG at z-level -7. Surprisingly, between-group analyses comparing the strength of correlation of the

PAG columns in the inhibitory and facilitatory group showed no significant differences between the two groups at rostrocaudal levels -5 and -7 between any of the columns.

Discussion

Our findings demonstrate brainstem differences in signal intensity related to CPM-evoked pain increases and decreases. Whilst hyperalgesic responses evoked such changes in a discrete set of brainstem structures – decreases in the LC, RVM and SRD as well as increases in the PAG, hypoalgesic responses involved the same structures with the addition of the A5 cell group. Indeed, a similar decrease in the LC and SRD is observed with a contrasting decrease in the PAG and increase in the RVM. Contradicting our hypothesis, the classic PAG-RVM circuit displayed signal intensity changes that correlated with both hypoalgesic (inhibitory) and hyperalgesic (facilitatory) responding. While the signal changes of the LC, A5 and SRD suggest they govern whether a CPM response has ‘inhibitor’ or ‘facilitator’ status, the PAG-RVM circuitry was only revealed when exploring relationships with pain intensity changes, suggesting a more modulatory role. Moreover, our data reveals that the lateral/dorsolateral PAG column is associated with CPM-mediated hypoalgesia while the ventrolateral/lateral PAG is involved in hyperalgesia.

Consistent with a previous human CPM imaging investigation¹⁷ which identified 43% of a pain-free sample exhibiting significant hypoalgesia during CPM, we found that 39% of pain-free participants experienced significant hypoalgesic CPM responses (inhibitors). Further categorizing individuals using permutation testing revealed that 25% of individuals experienced hyperalgesia upon application of the CPM paradigm (CPM facilitators), with the remaining 36% displaying no significant change in pain perception (non-responders). We found a smaller proportion of CPM facilitators than some previous studies which may be due to differing categorization or sub-grouping methods. Most commonly, the magnitude of CPM is calculated as the absolute or percentage change in pain intensity of a test stimulus between a control and CPM paradigm and any negative values represent CPM inhibition, positive values represent facilitation, and ‘0’ no CPM effect.^{42–44} Larsen and colleagues use a threshold of 5.3% change in pain intensity to determine response profiles resulting in 39% inhibitor, 42% facilitator and 19% non-responder sample composition.⁴⁵ However, a small pain intensity change may not be significantly different with statistical analysis requiring the acquisition of multiple test stimuli pain ratings. In this study we used permutation testing to determine at an individual level if pain intensity changes were indeed statistically significant and this might be the reason for our group proportions. A consensus on a preferred categorization protocol should be established for future studies.

Similar to previous investigations, an individual’s CPM response status was not significantly influenced by sex, age, test or conditioning stimuli intensities in our pain-free controls.^{46–48} Additionally, unlike other pain modulatory paradigms such as placebo analgesia, where it has been shown that the consistency of pain intensity ratings during repeated identical noxious stimuli is associated with the propensity to display a significant pain reduction,^{49,50} we show that baseline pain intensity rating variability is not related to subsequent CPM responses. This suggests a disconnect between expectation of forthcoming noxious inputs and CPM expression in healthy humans, and moreover, a behavioural delineation between CPM and alternate pain modulatory phenomena.

The canonical pain modulatory nuclei - PAG, RVM, SpV, were recruited during CPM and correlated with the magnitude of both hypoalgesic and hyperalgesic responses. Since preclinical studies demonstrated that PAG and RVM lesions did not alter DNIC responses and a 3 T fMRI study reported no CPM-related signal changes in these nuclei, it was thought these nuclei were not critical for CPM expression.^{12,13,17,51} This PAG and RVM only emerged when exploring linear

relationships between signal and pain changes suggesting these regions may act in concert to modulate the extent of pain intensity decrease (in the case of CPM inhibitors) or increase (in the case of CPM facilitators) rather than controlling whether or not a response occurs. The rostral location of these signal changes is consistent with a recent fMRI study that revealed a crude PAG somatotopic organization with noxious face stimulation activating the same rostrocaudal PAG level as that reported here, i.e. $Z = -5$, and body noxious stimuli activating caudal PAG.⁵² This finding raises the possibility that modulation of CPM responses by the PAG is also somatotopically organized. Interestingly, our data implicates discrete PAG columnar organisation in CPM expression. Indeed, we found lateral/dorsolateral PAG signal changes associated with CPM-evoked decreases in the case of CPM inhibitors, while the ventrolateral/lateral signal changes were associated with CPM-evoked increases.

We also report in humans the involvement of the A5 region during CPM, displaying both signal differences between inhibitory versus non-responder groups, and a positive correlation with the magnitude of hypoalgesia. A role for the A5 in CPM expression is consistent with previous preclinical studies that reported spinal application of an $\alpha 2$ -adrenoceptor antagonist preventing the inhibition of wide dynamic range dorsal horn neurons during activation of DNIC,⁵³ A5 activation inhibiting dorsal horn wide dynamic range neuronal firing, and A5 inhibition abolishing the DNIC response.¹⁵ Moreover, we also observed signal changes in the SRD during CPM-induced hypoalgesia, a finding consistent with a previous fMRI investigation.¹⁷ Preclinical studies have demonstrated that SRD lesions reduced the magnitude of C-fibre activity evoked by heterotopic noxious stimulation by 50%¹⁴ implicating its role in ‘triggering’ CPM responses. Similar to the RVM, the SRD appears to play a role in balancing pronociceptive and antinociceptive signalling⁵⁴ and our findings suggest reduced activity of the SRD shifts the balance towards inhibition.

The LC responded during both hypoalgesic and hyperalgesic responses. The LC has largely been considered a pain suppressor via noradrenergic action on $\alpha 1$ - and $\alpha 2$ -adrenoceptors in the spinal cord and SpV.^{55,56} Whilst LC lesions had no impact on DNIC responses,¹¹ a more recent study showed that optoactivation of the LC to spinal cord pathway abolished DNIC responses in rats,⁵⁷ complementing the reduced LC signal intensity changes in inhibitors. It has been posited that LC can evoke bidirectional (i.e. both inhibitory and facilitatory) effects on pain.⁵⁸ A more ventral portion of the LC exhibited signal intensity changes in CPM inhibitors compared to non-responders whilst a more dorsal portion exhibited changes in CPM facilitators. This complements a previous preclinical study whereby optoactivation of a ventral subgroup of LC neurons resulted in increased paw withdrawal to heat stimuli applied to the hindpaw of rats while that of the dorsal LC led to decreased thermal thresholds.⁵⁹

This study is not without limitations. Firstly, we did not account for finger movements during pain ratings. Since all participants moved the same fingers during both the control and CPM scans it is unlikely that movements would have influenced our results. Secondly, as the conditioning stimulus involved the induction of a relatively long-lasting muscle pain, our experimental design comprised a fixed-order, parallel CPM paradigm. Therefore, it was not possible to fully counterbalance the ordering of stimuli, and randomisation of control and CPM scans was not possible. Despite this, previous CPM studies have supported the use of fixed-order paradigms whereby application of the test stimulus alone precedes the CPM paradigm,⁶⁰ and no preference to the parallel or sequential nature of the CPM protocol.^{22,61} Extending from this, our study does not provide insights into the test-retest reliability of CPM although previous studies show inter- and intra-variation in the magnitude of the CPM effect as well as the categorization of individuals.^{62,63} This has been largely attributed to the components of experimental design including intensity, nature and combination of the test and conditioning stimuli, as well as the sites of stimulation, and parameter measures of the CPM effect such pain intensity changes and

thresholds.^{62–65} Such properties may contribute to the differences observed across neuroimaging CPM studies and hence the challenge of delineating the brain regions underlying CPM remains.

Thirdly, we divided participants into three groups based on pain intensity changes, ultimately reducing overall statistical power. However, whilst fMRI responses employed an initial threshold of $p < 0.005$ uncorrected for multiple comparisons with a cluster extent threshold of 4 contiguous voxels, critically we restricted our analysis to brainstem nuclei with well-described roles in pain modulation, increasing our confidence that the reported changes are real. The enhanced signal-to-noise afforded by 7 Tesla imaging allowed us to confirm that voxel peaks and boundaries were located within anatomically meaningful areas, further validating our findings. Furthermore, it is clear from our results that different brainstem circuitry underpins pain decreases and increases, thus not dividing individuals into response patterns is inappropriate and will not provide an accurate picture of the underlying brain circuitry.

It should also be noted that whilst we describe essentially two largely separate sets of brainstem structures, with some overlap, responding in inhibitor and facilitator groups, post-hoc analysis of signal changes for each region across all three groups revealed numerous regions in which there were no significant differences between inhibitors and facilitators, despite being significantly different to non-responders. This lack of difference between inhibitors and facilitators in some brainstem regions may suggest similar activation or inhibition of these nuclei during CPM with other regions being differentially activated to induce distinct responding. Alternatively, it may result from a lack of power given the necessity to categorise individual behavioural responses into three distinct groups. Importantly, this categorization of individuals into CPM inhibitor, facilitator and non-responder groups was essential for the identification of these brainstem networks.

With this limitation in mind, our data provides evidence of two overlapping networks, a pronociceptive network comprising the PAG, RVM, LC and SRD, and an antinociceptive network that differs with the additional involvement of the noradrenergic A5 during CPM expression. Alterations of these nuclei could also reflect a gating mechanism whereby these regions, in concert, regulate nociceptive processing and changes to determine whether an individual will exhibit no change, an inhibition or facilitation of pain perception induced by application of the CPM paradigm. These findings, in combination with those from pre-clinical studies, imply that the PAG and RVM may play a modulatory role influencing the degree of pain change whereas the LC, A5 and SRD play a permissive role associated with determining the status of the CPM response. Given that chronic pain is associated with altered pain modulatory function, future studies defining the precise brainstem circuitry responsible for such changes is important for a better understanding of chronic pain pathophysiology.

Study pre-registration

This study was not pre-registered.

Open material and methods

All materials, protocols, and analytical methods used in this study are openly available to ensure transparency and reproducibility. Detailed descriptions of experimental procedures, data collection instruments, and analysis pipelines are provided in the Materials and Methods section of the manuscript. Where applicable, supplementary documents include step-by-step protocols, parameter settings, and software versions.

Author contributions

Ashleigh H Wake: data curation, formal analysis, investigation, writing-original draft; Lewis S Crawford: data curation, formal analysis,

methodology, supervision, writing-review and editing; Noemi Meylakh; project administration, supervision, writing-review and editing; Alister Ramachandran: project administration, supervision, writing-review and editing; Vaughan G Macefield: project administration, supervision, writing-review and editing; Paul M Macey: project administration, methodology, software, validation, writing-review and editing; Tor Wager: conceptualisation, resources, writing-review and editing; Kirsty Bannister: conceptualisation, resources, writing-review and editing; Kevin A Keay: conceptualization, funding acquisition, methodology, resources, supervision, writing-review and editing; Luke A Henderson: conceptualization, data curation, formal analysis, funding acquisition, methodology, resources, supervision, writing-review and editing.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jpain.2026.106275](https://doi.org/10.1016/j.jpain.2026.106275).

Data availability

De-identified data can be made available upon reasonable request.

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